DEPI
Digital Egypt Pioneers Initiative
رواد مصر الرقمية

Track: Professional Data Analyst

RetailIQ

Online Retail Sales & Customer Analytics — from raw e-commerce transactions to next-day sales forecasts and a decision-ready dashboard.

- UK online retailer (UCI)
- 4-week delivery
- 5-person team

Business Problem

Sales swing with seasonality, demand, returns and country mix. Decision-makers lack reliable forecasts to plan inventory, marketing and customer retention.

Proposed Solution

An end-to-end analytics pipeline — clean, explore, forecast — that predicts next-day sales, estimates expected revenue and order volume, and delivers insights in Power BI.

Project Roadmap

SQL · Python · Power BI · DAX

WEEK 1 — Data Model & Cleaning
- Model entities, dimensions & measures
- Fix missing IDs, dupes, cancellations
- Clean & validate in SQL + Python
- Data dictionary & quality report

WEEK 2 — Analysis & KPIs
- Revenue, AOV, return & retention KPIs
- Trends by product, customer, country
- Top/bottom performers & segments
- Shortlist forecasting variables

WEEK 3 — Forecasting
- Predict next-day sales
- Compare models, validate metrics
- Estimate revenue & order volume
- Actual-vs-predicted, assumptions

WEEK 4 — Dashboard & Report
- Interactive Power BI + DAX model
- KPIs, trends, segments, forecast
- Management decision view
- Final report & presentation

Technology Stack

SQL · Python · pandas · NumPy · Matplotlib · scikit-learn · Power BI · DAX · Excel

Project Outcomes

- Interactive dashboard
- Next-day forecast
- Clean dataset + scripts
- Report & recommendations

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